



RUNBOOK

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

VCF 9 Fleet NTP Validation & Hardening

Validating NTP across every VCF component, finding config drift, and hardening it
— with the re-lock transient and the sos --ntp-health check.

v1.0

VMware Cloud Foundation 9

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INTERNAL

VCF 9 Fleet NTP Validation & Hardening

Prepared by: Virtual Control LLC **Date:** July 27, 2026 **Platform:** VMware Cloud Foundation 9.0.1

When to use. To validate NTP across every VCF component (time skew breaks SSO, certs, vSAN, and the cluster store), to find NTP **config drift**, and to harden it — all self-service.

1. Validate Every Component (read-only)

ESXi hosts (SSH as root):

```
esxcli system ntp get          # Enabled: true, Time Synchronized: true, Servers: ...
ntpq -pn                      # want:  *<ntp-ip>  ... reach 377  (the * = locked)
```

vCenter (API):

```
TOK=$(curl -sk -u 'administrator@vsphere.local:<pw>' -X POST https://vcenter.lab.local/api/session | tr -d ' ')
curl -sk -H "vmware-api-session-id: $TOK" https://vcenter.lab.local/api/appliance/ntp          # servers
curl -sk -H "vmware-api-session-id: $TOK" https://vcenter.lab.local/api/appliance/timesync    # want
"NTP"
curl -sk -H "vmware-api-session-id: $TOK" https://vcenter.lab.local/api/appliance/system/time  # current
time
```

NSX (API): `curl -sk -u 'admin:<pw>' https://nsx-vip.lab.local/api/v1/node/services/ntp`

SDDC Manager (SSH as `vcf`): `timedatectl` → System clock synchronized: yes ; `ntpq -pn` → `*<ntp-ip> reach 377` .
(SDDC NTP config via API: `GET /v1/system/ntp-configuration` .)

Read the signals right: - `reach 377` (octal, all 8 recent polls good) + a `*` on the peer = fully locked. Lower reach (3/7/17...) = still re-locking. - Offsets in **milliseconds** are healthy; the fleet should agree with each other (compare each component's clock). - On SDDC Manager, **NTP service: inactive in `timedatectl` is NOT a fault** — that line tracks `systemd-timesyncd`, which is off because `ntpd` is the active daemon. Trust `System clock synchronized: yes` + `ntpq` .

2. Find Config Drift

List each component's configured NTP server. Drift to watch for: - Some components reference the source by **IP**, others by **FQDN**, others by **both**. - **Hostname-only on any appliance is a latent risk** — a DNS hiccup can desync it

while IP-based components stay fine. (Verified 2026-07-27: SDDC Manager was hostname-only.)

3. Harden — SDDC Manager NTP to IP + FQDN

Set the fleet NTP config (SDDC pushes it to all components) to include **both** the IP and the FQDN of the source: - **VCF Operations / SDDC:** update the NTP configuration to list both `<ip>` and `<fqdn>`. - API form: `PUT /v1/system/ntp-configuration` with both servers.

Expect a transient "not synchronized" window after any NTP change. The fleet-wide push **restarts ntpd on every component**, so each drops sync and re-locks over **~8 minutes** (`reach` climbs 3→7→17→...→377). During that window **NTP health-check validations will log Failed** — that is the re-lock window, not a real failure. **Wait ~8 minutes, then re-validate.**

4. Re-Validate After the Change

- 1. Wait ~8 minutes from the push.
- 2. Confirm every component: `Time Synchronized: true / synchronized: yes , *<ip> peer, reach 377`.
- 3. Run the official NTP health check on SDDC Manager (as `vcf` , `sudo`): `bash sudo /opt/vmware/sddc-support/sos --ntp-health` **Note the flag is `--ntp-health`** (verified 2026-07-27 — *not* `--ntp-check`). Exit 0 = the check ran; read per-component results in the health-check report dir it prints.

Best practice notes. - **IP + FQDN of the same server is not redundancy** — it's one box under two names (the IP entry removes the DNS dependency; the FQDN adds nothing). For *real* NTP resilience, add a **genuinely different second time source**. - A workstation clock being seconds off the fleet is a **local** issue, not a VCF one — judge the fleet by internal consistency + `reach 377` , not by a single admin PC's clock.

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